PRNN Programming Assignment 1

Arjun Mallick
Sr. 25989

1.1 Gradient Descent vs Closed Form

- **Closed Form Solution:** $w_{CF} = \begin{bmatrix} 4.1929 & 2.4719 \end{bmatrix}$
- **Batch GD Solution:** $w_{GD} = \begin{bmatrix} 4.1926 & 2.30535 \end{bmatrix}$

L2 Norm Difference Between Weight Vectors:

$$\|w_{GD} - w_{CF}\|_2 = 0.16655$$

1.2 Primal to Dual Translation

The hard-margin SVM assumes that the dataset is perfectly linearly separable, meaning there must exist a hyperplane satisfying $y_i(w^T x_i + b) \geq 1$ for all training points. Since Dataset 1 is not linearly separable, these constraints cannot be satisfied simultaneously, and therefore a valid hard-margin solution with well-defined support vectors **does not exist**.

2.3 The Singularity Trap

The matrix $X^T X$ is numerically ill-conditioned due to extreme multicollinearity ($D > N$), giving a condition number of 1.27386×10^{19}, which makes the closed-form OLS inversion unstable even though it **does not raise a Python exception**.

Condition Number vs λ: The relationship between the regularization strength and numerical stability is shown in Fig. 1, where increasing λ reduces the condition number.

2.4 Lasso Sparsity

Regularization Path: Fig. 2 illustrates how LASSO progressively drives feature weights to zero as the regularization strength λ increases.

λ where 50% weights are zero: 0.37276 (Sparsity obtained $\approx 50.6\%$)

3.5 Vectorized Logistic Regression

Learning Curves: Fig. 3 compares convergence behavior under a stable learning rate and an excessively large learning rate. When the Lipschitz condition is violated, the learning curve oscillates because the step size exceeds the smoothness bound of the logistic loss gradient, causing gradient descent to overshoot the optimum.

3.6 GMM Initialization and One EM Iteration

Initial Log Likelihood: -3115871.9362
Log Likelihood After One Iteration: -2512521.1361

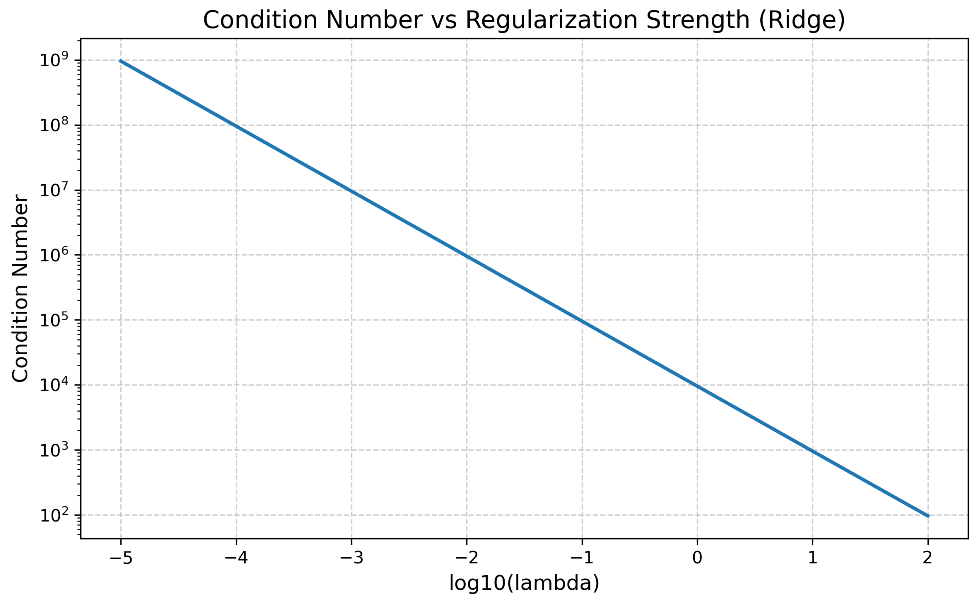

Figure 1: Condition number of $(X^T X + \lambda I)$ as a function of λ (log scale), showing improved numerical stability with increasing regularization.

3.7 EM Convergence and Crashes

Log Likelihood vs Iterations: Fig. 4 shows the evolution of log-likelihood across iterations for GMM models with $K \in \{2, 4, 6\}$, illustrating the monotonic convergence property of the EM algorithm.

Python Error when covariance matrix forced to zero variance:
`LinAlgError: When allow_singular is False, the input matrix must be symmetric positive definite`

3.8 Decision Boundaries

Fig. 5 shows the learned Gaussian components and the resulting decision boundaries when the GMM is trained using the first two feature dimensions.

3.9 SVM with Slack KKT Verification

The soft-margin SVM objective was solved using a subset of the dataset (size 1000). The following results illustrate examples of points satisfying different KKT conditions.
Inside Margin: No points detected inside the margin.
On Margin: No points detected exactly on the margin boundary.
Outside Margin:

```
Index: 6226
Mu: 0.00
```

3.10 Mercer Condition

The eigenvalues of the RBF kernel matrix are all non-negative, confirming that the kernel matrix is positive semi-definite and therefore satisfies Mercer's condition.
Subset Shape: $(1000, 50)$
Kernel Matrix Shape: $(1000, 1000)$
Minimum Eigenvalue: 0.9999999992399248

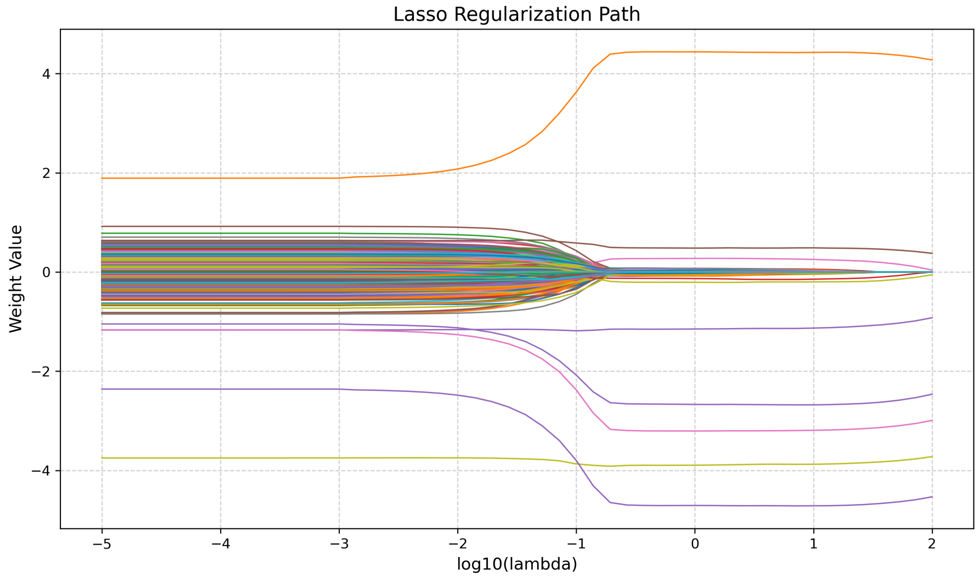

Figure 2: Regularization path of LASSO showing coefficient shrinkage and sparsity as λ increases.

3.11 Hyperparameter Topography

Fig. 6 shows the validation accuracy across the grid of hyperparameters (C, σ) for the RBF SVM.

Most Severe Overfitting:

$$(C, \sigma) = (1.0, 0.001)$$

$$(train\_acc, val\_acc) = (1.0, 0.495)$$

4.12 Multiclass Extension and Scaling

Iterations to Converge (Raw Data): 1000 (maximum iterations reached)

Iterations to Converge (Standardized Data): 257

4.13 Vectorization Benchmarking

- **Loop Implementation Time:** 2.2227 seconds
- **Vectorized Implementation Time:** 0.0422 seconds
- **Speedup Factor:** 52.69×

4.14 The Curse of Scale

- **Validation Accuracy (Raw):** 0.8801
- **Validation Accuracy (Standardized):** 0.9217
- **Accuracy Difference:** 0.0415

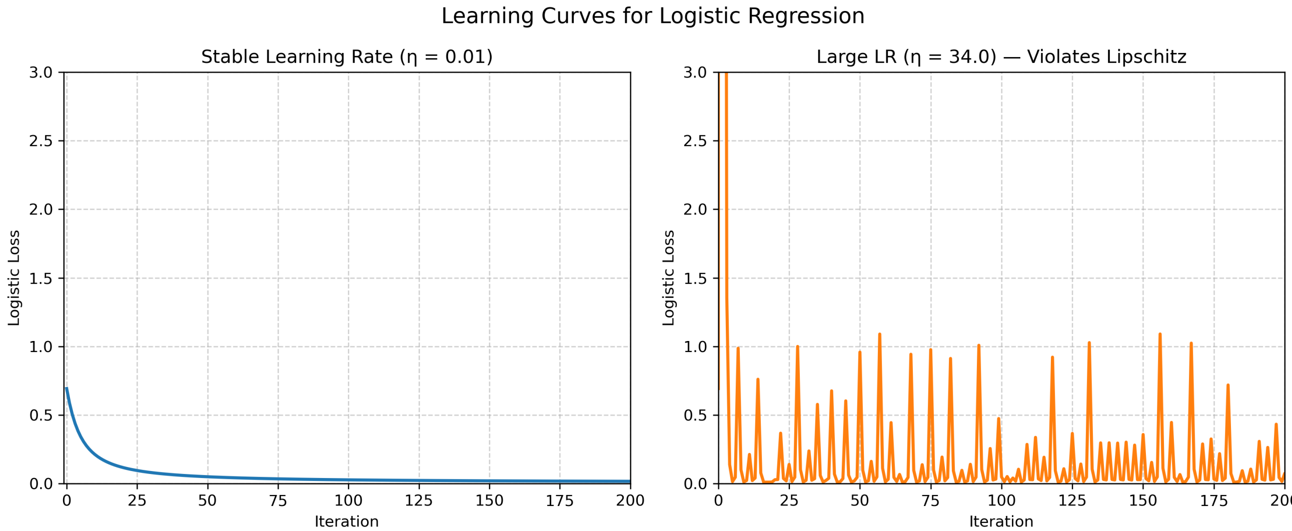

Figure 3: Learning curves for logistic regression showing stable convergence for a small learning rate and oscillatory behavior when the Lipschitz condition is violated.

4.15 Bayes Optimal and Underflow

Feature Dimension where Underflow Occurs:

$$D = 28$$

5.16 Empirical Bias Variance via Bootstrapping

Metric	**Linear Model ($d = 1$)**	**Polynomial Model ($d = 15$)**
$Bias^2$	2.26520	2.25336
$Variance$	0.00108	0.35626

Table 1: Empirical bias and variance estimates obtained using bootstrap resampling.

5.17 Frequentist vs Bayesian Uncertainty

Frequentist Variance of Slope: 1.6557×10^{-5}
Posterior Variance: 1.3942×10^{-5}
Distribution Plot: Fig. 7 compares the frequentist sampling distribution of the slope estimator with the Bayesian posterior distribution of the slope parameter.

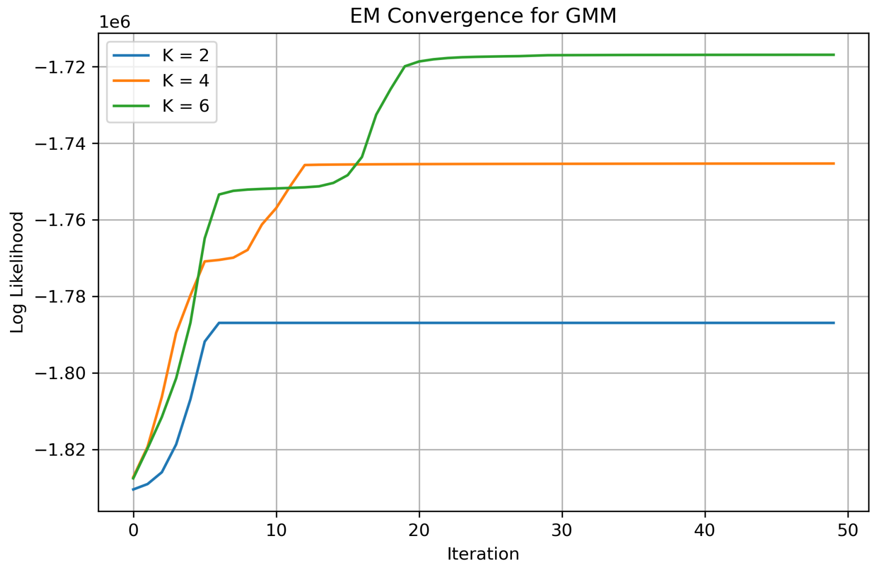

Figure 4: Log-likelihood progression during EM training for Gaussian Mixture Models with different numbers of components.

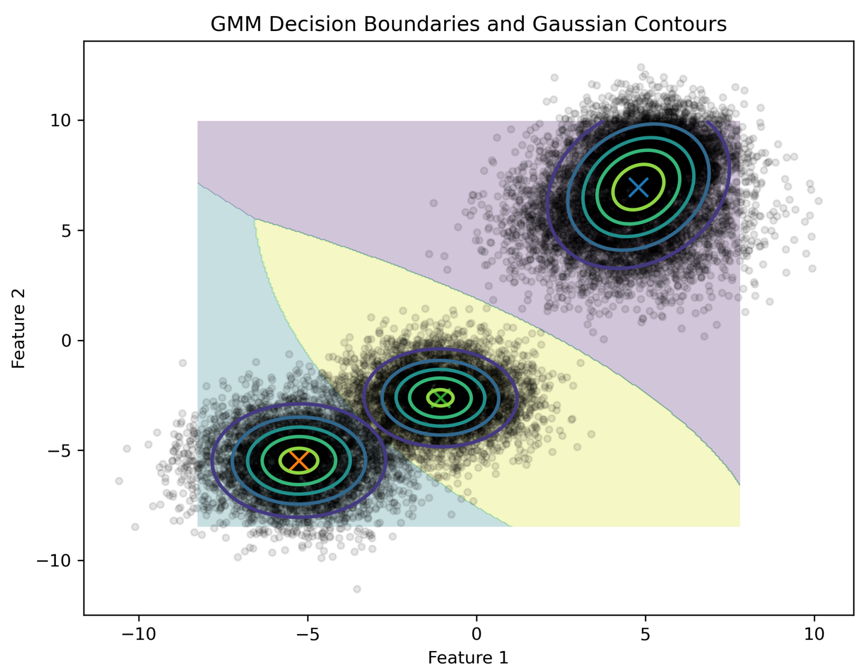

Figure 5: Decision boundaries and covariance contours learned by a GMM trained on the first two dimensions of the dataset.

"continuing from previous page"

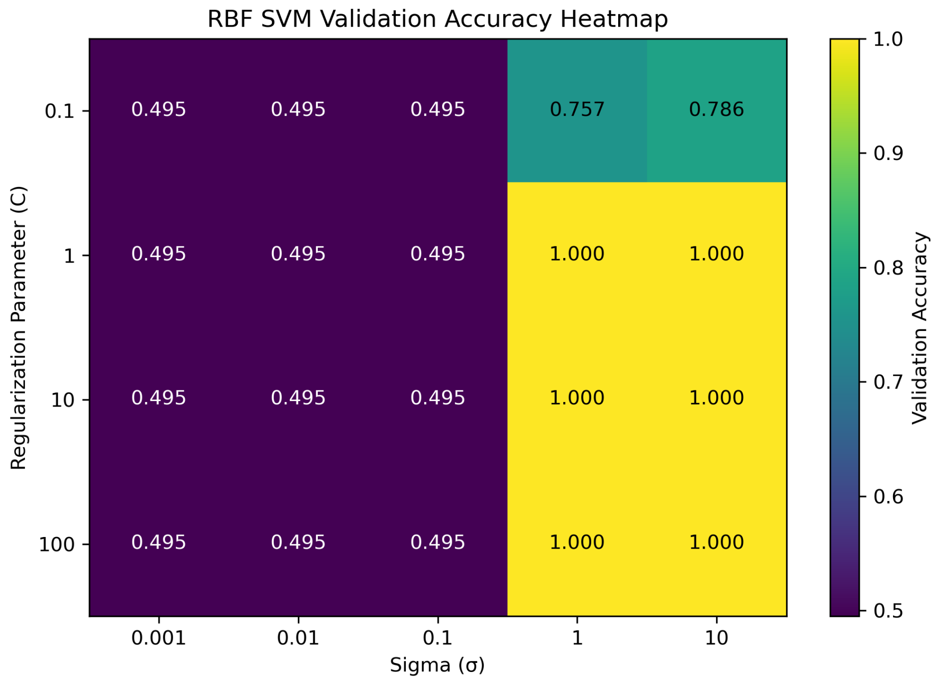

Figure 6: Validation accuracy heatmap over the (C, σ) hyperparameter grid for the RBF SVM.

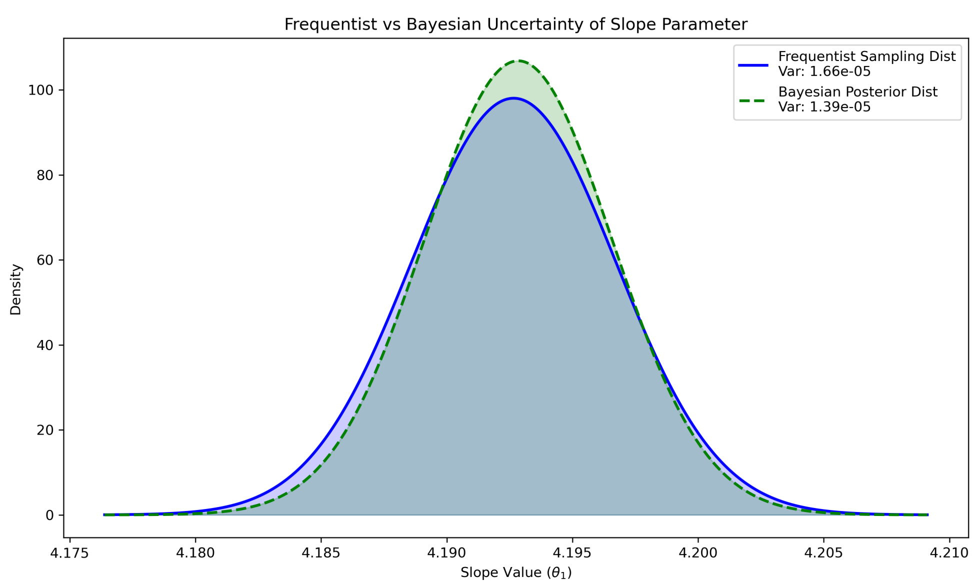

Figure 7: Comparison of frequentist sampling distribution and Bayesian posterior distribution for the slope parameter.